Worksheet 5.4

Precipitation

Student: .... Class: ....

Observing precipitation

1. The following photograph shows the result of adding a colourless solution of lead nitrate to a colourless solution of potassium iodide in a test tube.

Photo: Lindsey Moore/ Shutterstock.com

(a) A yellow precipitate forms. Define the term 'precipitate'.

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(b) The beaker is allowed to stand for 1 hour. What will happen to the precipitate in that time?

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(c) Name the compound making up the precipitate.

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(d) Write a word equation for the reaction.

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Predicting precipitation

2. The following table provides information on the solubility of various salts in water.

Anion	Soluble	Slightly soluble	Insoluble
Nitrate	All	None	None
Chloride	Most	$PbCl_2$	$AgCl$
Sulfate	Most	$CaSO_4$, Ag_2SO_4	$BaSO_4$, $PbSO_4$
Carbonate	Na_2CO_3, K_2CO_3, $(NH_4)_2CO_3$	None	Most
Hydroxide	$NaOH$, KOH, NH_4OH, $Ba(OH)_2$	$Ca(OH)_2$	Most

Use this information to predict whether a precipitate will form when the nominated solutions are mixed. If a precipitate forms, name the precipitate.

(a) Lead nitrate solution + sodium sulfate solution.

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(b) Copper(II) sulfate solution + potassium hydroxide solution.

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(c) Barium chloride solution + calcium nitrate solution.

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